

Generative Reference Models for Spatial Entropy Traces in IoT Sensor Fault Detection

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Abstract—We test whether local correlation-spectrum entropy trajectories improve sensor-group monitoring when compared with a conditional joint normal reference. The formulation combines entropy level and change, explicit missing-data abstention, spatial scan calibration and inspectable sensor participation. A compact graph-conditioned diffusion model and a context-only block bootstrap supply matched entropy, synchronization and combined comparisons; full correlation matrices, raw residuals, CUSUM and a GDN adaptation are also evaluated. One synthetic benchmark and semi-synthetic evaluation on Intel Lab and PEMS-BAY recordings provide 360 scheduled fault injections in 56 recording blocks, with three training seeds. At nominal per-issuance $\alpha = .10$, diffusion entropy recall is 0.326/0.106/0.139, versus 0.315/0.250/0.181 for synchronization, in dataset order. The paired comparisons do not establish a consistent entropy advantage in both detection and localization. Untouched entropy exceedance ranges from 0.032 to 0.308, so equal nominal budgets do not establish equal realized false-alert rates. Analytical bounds and an exact matched-correlation counterexample delimit the information captured by entropy. Saved predictions, calibrated diagnostics and a replay dashboard make the resulting attention signals auditable without claiming verified field failures or causal explanations.

Index Terms—sensor monitoring, digital twins, data spaces, spectral entropy, conditional diffusion

I. INTRODUCTION

Sensor-fed digital twins depend on measurements whose collective behavior can change before a single extreme value becomes conspicuous. A copied stream may make several devices unusually alike; independent interference may break a previously shared pattern. A legitimate operating transition can also change coordination. A monitoring layer for data spaces should identify the affected measurements, retain their provenance and expose this ambiguity rather than equate a change with a verified physical failure.

We investigate an informational description of this behavior: the entropy of the correlation spectrum within spatially selected sensor groups. A spectrum concentrated in a few modes has low entropy; broadly distributed modes have high entropy. Monitoring both directions can reveal different reorganizations. However, direction alone is insufficient: expected seasonality, window length, missing observations and changes in normal collective activity also affect the trace. The question is whether comparing local entropy trajectories with a conditional joint normal reference improves detection and localization at a matched calibration budget, relative to non-entropic synchronization traces.

The motivating connection to Mantzaris’s research is specific. Entropy trajectories have been used to describe evolving segregation under different network topologies [1] and collective

flocking organization [2]. We transfer the use of evolving entropy as a description of collective organization, and the attention to the network that defines interactions. The former study uses degree and homophily distributions; the latter constructs thermodynamic quantities from a flocking model. Here the probability vector consists of normalized eigenvalues of a *sensor correlation matrix*. This is not a degree-distribution entropy, a thermodynamic entropy or a causal explanation. We establish no temperature, heat flow or second-law claim.

Our contribution is a reproducible formulation and empirical test of local level-and-change scores, historically issued generative reference blocks, explicit quality abstention, scan calibration and inspectable localization. The study separates feature choice from reference-model choice in a factorial comparison, includes a full-correlation detector, and preserves negative findings. The output is an attention signal with observed evidence; injected faults, legitimate controls and unknown native causes remain distinct.

II. RELATED WORK

Spectral entropy is established. Effective rank uses the exponential entropy of normalized singular values [3]; for a PSD correlation matrix, singular values equal eigenvalues. Graph-entropy monitoring also predates this study: FINGER approximates von Neumann entropy of graph Laplacians and applies graph distances to anomaly detection [4]. Our correlation spectrum differs from a Laplacian spectrum, and its spatial meaning comes from group selection, not an ordering of sensors inside the matrix.

Diffusion forecasting and imputation are likewise prior tools. TimeGrad models multivariate predictive distributions [5], CSDI conditions joint diffusion on masked observations [6], and DiffSTG incorporates spatiotemporal graph structure [7]. DiffAD [8], ImDiffusion [9] and ICDiffAD [10] demonstrate that diffusion-based time-series anomaly detection is not new. Our information boundary is a forecasting boundary: the main generator receives no observed value from its target block. We ask about the resulting distributions of collective measurements, rather than claiming a new denoising architecture.

GDN learns a sensor graph and forecasts values to detect departures from inter-sensor relationships [11]. It motivates our strong graph baseline. Conformal work with dependent observations explicitly addresses assumptions needed beyond exchangeability [12]. Our simpler scan-max rank calibration is an application of established rank reasoning, with empirical time-series diagnostics, not a new distribution-free stream theorem.

III. LOCAL MEASUREMENTS AND NORMAL REFERENCES

A. Spatial groups and data quality

Let $G = (V, E)$ be a fixed supplied physical or proximity graph. A predeclared family $\mathcal{G}$ contains deduplicated neighborhoods of sizes 6, 12 and 24 around 24 fixed index-spaced centers. Unweighted graph-hop distance selects membership; geographic distance breaks finite ties, with node-order fallback for unreachable nodes when a component is smaller than the requested group. Windows $W \in \{24, 48, 96\}$ and lags $d = W/4$ are scanned only when $W \geq 2|g|$. Channels are processed separately: temperature and humidity never enter the same correlation matrix.

For one group/channel window, retain rows observed at every sensor. With m columns and n retained rows, require $n \geq \max(2m, \lceil .8W \rceil)$. Otherwise abstain. Variance below 10^{-8} in training-scaled units gives a separate flatline flag. It does not receive an entropy of zero. Neural inputs use masked zero after training-only scaling, while feature calculation retains the original missingness. Generated samples receive the observed window mask *after* generation to compare equal measurement support.

B. Entropy and synchronization

Center and standardize each retained column by its sample standard deviation, forming $Z \in \mathbb{R}^{n \times m}$. Define

$$\mathbf{R}_{g,t} = Z^\top Z / (n - 1), \quad \tilde{\mathbf{R}}_{g,t} = (1 - \lambda)\mathbf{R}_{g,t} + \lambda I, \quad (1)$$

$$p_k = \frac{\lambda_k(\tilde{\mathbf{R}}_{g,t})}{\text{tr}(\tilde{\mathbf{R}}_{g,t})}, \quad H_{g,t} = -\frac{\sum_k p_k \log p_k}{\log m}, \quad (2)$$

$$\Delta H_{g,t} = H_{g,t} - H_{g,t-d}. \quad (3)$$

We use $0 \log 0 = 0$, $\lambda = .05$, a symmetric eigensolver and a roundoff tolerance of $64m$ machine epsilons. Materially negative eigenvalues are an error. The Gram product disables reduced-mantissa TF32; the network can use mixed precision. CPU float64 cases check CUDA float32 measurements.

The non-entropic features are mean signed correlation C , mean absolute correlation C_{abs} , pairwise squared disagreement D , and leading concentration $P = \lambda_{\max}(\mathbf{R})/m$, each with a lagged change. With our within-window sample normalization,

$$D = \frac{1}{n \binom{m}{2}} \sum_{s,i < j} (z_{si} - z_{sj})^2 = 2 \frac{n-1}{n} (1 - C). \quad (4)$$

Thus D is redundant with C at fixed n . Low H can reflect positive *or* negative dependence; displayed pairwise signs are retained.

C. Joint conditional diffusion

For the score at t , the generated block spans $t - 119$ through t ; it contains both windows for the largest $W + d = 120$. The 48-row conditioning history ends at $t - 120$, strictly before the target. Smaller windows use suffixes of the same block, making their longer forecast lead explicit. Context includes measurements, masks, fixed coordinates/graph and known calendar harmonics.

Our compact independent implementation follows the graph-temporal conditional denoising ideas of DiffSTG/CSDI. A history encoder, coordinate/calendar embeddings, width-24 gated temporal convolutions and three graph aggregation blocks predict joint noise across all sensors. It is not a reproduction of DiffSTG's UGnet. For noise step s ,

$$Y_s = \sqrt{\alpha_s} Y + \sqrt{1 - \alpha_s} \epsilon, \quad (5)$$

$$\mathcal{L} = \mathbb{E} \left[\frac{\|M_{\text{loss}} \odot (\epsilon - \epsilon_\theta(Y_s, s, C, G))\|_F^2}{\sum M_{\text{loss}}} \right]. \quad (6)$$

Only valid training targets enter the loss. The cosine schedule [13] uses 100 steps; compatible DDIM [14] uses 20 inference steps, independent initial noises and $\eta = 0$. Denoised states are clipped to $[-8, 8]$ in training-scaled units. Three seeds, Adam at 10^{-3} , batch 8 and at most 2400 updates are used; development checks every 200 updates select checkpoints, with patience four. The main sample count is 64, chunked by 16.

The bootstrap reference samples intact joint training blocks from 64 candidate donors selected by historical context and calendar similarity. Selection sees no target values. This preserves empirical temporal/cross-sensor organization and provides the same feature-by-reference factorial comparison. Neither model identifies causal counterfactuals of real failures.

D. Trajectory scores, calibration and localization

For $u = (H, \Delta H)$, generated samples provide component-wise median c_j and scale $s_j = 1.4826 \text{ median}_b |u_j^{(b)} - c_j| + \epsilon_j$. The floor is 5% of the development IQR, with numerical minimum 10^{-4} , then frozen. The proposed bidirectional trajectory score is

$$A_{g,t}^H = \max_{j \in \{1,2\}} \frac{|u_{g,t,j} - c_{g,t,j}|}{s_{g,t,j}}. \quad (7)$$

MAD scaling implies no Gaussian tail model. Synchronization and combined scores use the same operation on their declared feature components. At least 80% of reference samples must support a feature. Independent comparisons include Frobenius distance to the sample-mean full correlation matrix, raw-value predictive residuals, a restarted finite-memory two-sided CUSUM and a labeled GDN reimplement. GDN uses causal observed target history for one-step prediction, an advantage over the longer reference forecast; its learned top- k graph and graph attention are retained, while its head, normalization and group mapping are explicit adaptations. Quality hybrids are reported separately from entropy-only detection.

The scan maximum $M_t = \max_{g \in \mathcal{G}} A_{g,t}$ includes every declared group, channel and eligible window. All-abstained units have $M_t = -\infty$. Disjoint context-plus-target calibration units, separated by 168 observations, produce $M_1, \dots, M_n$. We use

$$p_t = \frac{1 + \sum_{b=1}^n \mathbb{I}\{M_b \geq M_t\}}{n + 1}. \quad (8)$$

Displayed group values compare $A_{g,t}$ with the same calibration maxima and are labeled scan-adjusted. Neither p nor $1 - p$ is a

fault probability. The statement is per issuance, never indefinite stream-wide control.

Sensor participation averages group scores with inverse-size weights. Diffusion-H IoU on six development injections selects a shared budget in $\{6, 12, 24\}$ for the highest-ranked sensors; participation is not causal attribution. Oracle overlap measures the best single candidate group, not an upper bound on combined participation. Display merging uses group Jaccard overlap at least .5, retaining original records.

IV. PROPERTIES AND LIMITS

Proposition 1. *For PSD correlation matrices, $0 \leq H \leq 1$. At zero shrinkage, identity and rank-one matrices attain 1 and 0. With positive shrinkage the minimum is the entropy of $(1 - \lambda + \lambda/m, \lambda/m, \dots, \lambda/m)$.*

Proof. The nonnegative eigenvalues sum to m , hence define a probability vector. Shannon’s bound gives the first statement. Shrinkage gives every probability at least λ/m ; the displayed most-concentrated vector majorizes all feasible vectors, and entropy is Schur-concave. $\square$

For n centered observations, rank is at most $\min(m, n - 1)$, so zero-shrinkage entropy is at most $\log \min(m, n - 1) / \log m$. Autocorrelation further reduces effective information without changing this algebraic ceiling; normalization does not equalize finite-sample distributions across groups or windows.

Proposition 2. *For $\mathbf{R}(\rho) = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$, $0 \leq \rho < 1$,*

$$\frac{dH}{d\rho} = \frac{m - 1}{m \log m} \log \frac{1 - \rho}{1 + (m - 1)\rho} \leq 0. \quad (9)$$

Proof. The eigenvalues are $1 + (m - 1)\rho$ along $\mathbf{1}$ and $1 - \rho$ on its orthogonal complement. Differentiate their normalized entropy; constant derivative terms cancel because the probabilities sum to one. For shrinkage, replace ρ by $(1 - \lambda)\rho$ and multiply by $1 - \lambda$. $\square$

This is a statement about equicorrelation, not a universal law of faults.

For $m = 4$, equicorrelation .2 has spectrum $(1.6, .8, .8, .8)$; two separate correlated pairs at .6 have spectrum $(1.6, 1.6, .4, .4)$. Both matrices are positive definite, with mean pairwise correlation .2 and leading concentration .4. Yet their zero-shrinkage entropies are .960964 and .860964 (Fig. 2). Entropy distinguishes these organizations lost by those simple summaries. It cannot contain more information than full $\mathbf{R}$.

Proposition 3. *For an exchangeable set of calibration and null-test maxima from a fixed scoring rule, (8) satisfies $\Pr(p \leq \alpha) \leq \alpha$.*

Proof. Under random tie breaking, symmetry makes the test rank among $n + 1$ scores uniform. Greater-or-equal tie counting is conservative. Thus the exceedance probability is at most $\lfloor \alpha(n + 1) \rfloor / (n + 1)$. $\square$

Neither independent simulation seeds nor gaps establish exchangeability: synthetic calendar phases differ across partitions, and real blocks remain dependent. Distribution shift and

contamination also require empirical evaluation. The minimum attainable p is $1/(n + 1)$.

Counterexamples delimit interpretation. Diagonal sign changes $\mathbf{SRS}$ and fixed-group permutations preserve the spectrum. Offsets and positive rescaling preserve correlations. A coherent fault and a legitimate common driver can share H . Flatlines make correlations undefined. The detector therefore needs signed correlation evidence and separately evaluated raw/quality alerts.

V. DATA AND PREREGISTERED EVALUATION

The synthetic benchmark uses random geometric graphs and a stable process $x_{t+1} = (.82\mathbf{I} - .25L)x_t + Bf_t + .12\epsilon_t$, with L divided by maximum degree, documented daily/weekly signals, spatial factor loadings and latent OU drivers. Spectral radii are checked below one. Entire simulations have disjoint train, development, calibration and test seeds; the three node counts are configurations of one benchmark, not three primary datasets.

Intel Lab [15] retains timestamps, masks, coordinates and units. Right-labeled two-minute bins take the last observed whole row, with no interpolation. A coverage-only audit, before Intel training/test scoring, excludes the terminal sustained availability collapse after March 23; the complete recording is retained for the quality sensitivity. All 54 motes remain. Primary channels are temperature and humidity, with separately audited light/voltage. PEMS-BAY uses the original DCRNN release [16], its road distances and all 325 sensors. Its actual file spans January–June 2017; the one-hour March 12 gap is explicitly masked. Zero values follow the release’s missing-value evaluation convention. Per-value upstream imputation lineage is unavailable, a limitation of using the processed release.

Real recordings use chronological 60/15/10/15% splits. Windows never cross split or synthetic-episode boundaries. Median/IQR scaling is training-only. Training-loss screening excludes standardized magnitudes above eight; an unscreened sensitivity tests this choice. Unlabeled real training recordings are nominal, not certified healthy.

The protocol was committed before full runs. All methods see identical controlled injections into untouched held-out recordings. The real-data study is *semi-synthetic*. Twelve fault mechanisms span copying/replay collapse, common interference, coupling loss/decorrelation, noise, delays, gain/offset drift, flatline/dropout, anticorrelation, mixed changes, matched mean correlation and marginal-preserving rearrangement. Severities .6/1, durations 12/48/96 and connected/partly disconnected groups are balanced in a fixed 72-event grid. Legitimate synchronized transitions and untouched recordings are controls. Entropy directions are measured, not assigned from the mechanism name. Whitening/recoloring matches sample mean correlation and marginal moments when full rank. Short rank-deficient cases use a Gaussian population target and do not guarantee matching realized correlation. All scheduled injections remain in the denominators; a saved audit flags unobservable interventions. For direction strata, take the median

Historically issued forecast: context ends before the entire observed target

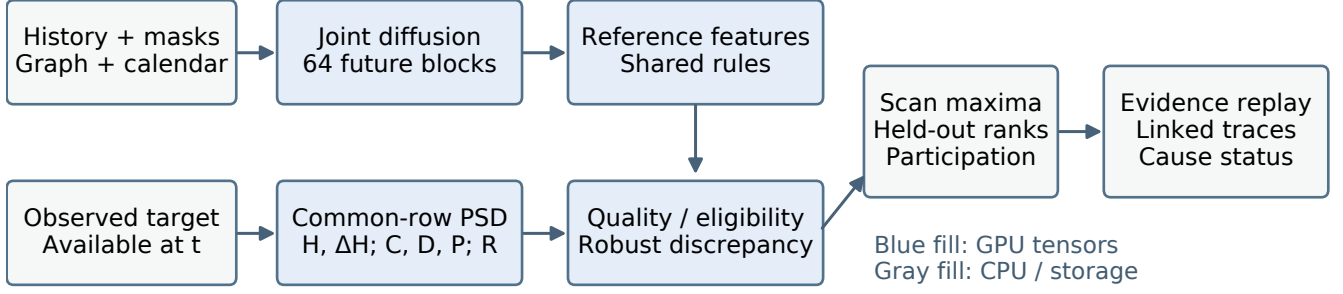


Fig. 1. Historically issued joint references and observed blocks share the same group/channel measurement rules. Missingness is applied to predictions only for measurement comparison. GPU stages are shaded; data parsing, manifests, summaries and replay rendering run on CPU.

TABLE I

PRIMARY DATASETS AND FROZEN SPLITS. COUNTS ARE TIME ROWS; INDEPENDENT SYNTHETIC EPISODES DEFINE THE PARTITIONS. INTEL AND PEMS FAULTS ARE CONTROLLED INJECTIONS, NOT CONFIRMED FIELD FAILURES.

Dataset	Nodes	Train	Dev.	Cal. / test	Interval; primary channels	Source span used
Synthetic (3 configs)	64/128/256	49,152	12,288	8,192 / 12,288	5 min; signal	independent simulations
Intel Lab	54	10,782	2,695	1,797 / 2,696	2 min; temp., humidity	28 Feb.–23 Mar. 2004
PEMS-BAY	325	31,276	7,820	5,212 / 7,820	5 min; speed (mph)	1 Jan.–30 June 2017

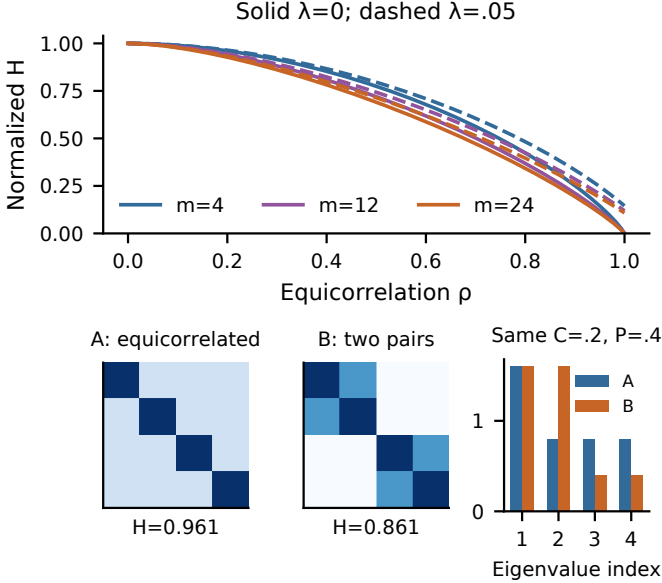


Fig. 2. Analytical entropy curves and the exact matched-summary counterexample; these are mathematical illustrations, not measured detection performance.

within-event ΔH in the predefined group with greatest overlap with the injected sensor set; truth is used only for evaluation. A 10^{-4} dead band labels stable traces. The sign-gated ablation uses strictly positive/negative observed changes, independent of truth, and recalibrates its scan.

The primary scan budget is $\alpha = .10$, with $.05/.20$ reported where attainable. Alerts are consecutive flagged issuance runs, with stride 12. Their onsets must match true event intervals one-to-one, with zero trailing tolerance; pre-existing alarms

receive no credit. No point adjustment is used. Event PR area integrates the upper precision envelope over attainable one-to-one recalls and retains the raw curve. Localization scores missed events as zero, with conditional detected-event results also available. Delays include window and stride effects. Table delays additionally include the measured mean complete comparison-pipeline latency under conservative serial execution; model-only and dedicated entropy-pipeline times are also reported separately.

Resampling uses whole independent synthetic simulations or disjoint real recording blocks, retaining all injections and averaging training seeds within each block. We report paired 95% bootstrap intervals, type macro scores, prevalence and missed counts. Real-block intervals are descriptive: gaps do not establish independence, and Intel has only eight blocks. Overlapping windows are not replicates. Native background alerts are not verified false physical-fault detections.

VI. RESULTS

A. Detection, localization and the feature hypothesis

The full primary study contains 360 scheduled fault injections within 56 recording blocks. All three seeds are complete; the three synthetic node counts remain configurations of one benchmark. Table II gives the reference-by-feature comparison and required baselines. Figure 3 includes whole-block uncertainty and the measured direction strata. The paired comparisons do not establish a consistent entropy advantage in both detection and localization.

The mean fraction of eligible diffusion-entropy group windows is $0.979/0.039/0.962$ in dataset order. Intel performance is therefore dominated by limited sample support. The shared

TABLE II

MAIN RESULTS AT NOMINAL $\alpha = .10$. EACH CELL IS RECALL / IOU / EVENT-PR AREA / MEDIAN DETECTED-EVENT DELAY (MINUTES). DELAY ADDS MEASURED MEAN COMPLETE COMPARISON-PIPELINE LATENCY TO THE OBSERVATION TIMESTAMP, A CONSERVATIVE SERIAL EXECUTION CONVENTION. LOCALIZATION COUNTS MISSES AS ZERO. VALUES SUMMARIZE THREE SEEDS; SYNTHETIC CONFIGURATIONS FORM ONE DATASET. EQUAL NOMINAL BUDGETS DO NOT IMPLY EQUAL REALIZED BACKGROUND RATES.

Method	Synthetic	Intel Lab	PEMS-BAY
Bootstrap H	0.188 / 0.013 / 0.151 / 175.0	0.208 / 0.027 / 0.049 / 70.0	0.509 / 0.029 / 0.134 / 55.0
Bootstrap S	0.196 / 0.021 / 0.159 / 175.0	0.185 / 0.028 / 0.028 / 70.0	0.403 / 0.026 / 0.165 / 115.0
Bootstrap H+S	0.196 / 0.021 / 0.159 / 175.0	0.185 / 0.028 / 0.028 / 70.0	0.398 / 0.028 / 0.151 / 115.0
Diffusion H	0.326 / 0.017 / 0.139 / 175.0	0.106 / 0.009 / 0.036 / 70.0	0.139 / 0.006 / 0.150 / 175.0
Diffusion S	0.315 / 0.021 / 0.134 / 175.0	0.250 / 0.031 / 0.045 / 46.0	0.181 / 0.005 / 0.141 / 175.0
Diffusion H+S	0.315 / 0.021 / 0.134 / 175.0	0.255 / 0.027 / 0.052 / 70.0	0.116 / 0.006 / 0.153 / 175.0
Diffusion full R	0.338 / 0.026 / 0.132 / 175.0	0.176 / 0.018 / 0.024 / 70.0	0.162 / 0.007 / 0.144 / 175.0
Diffusion raw	0.054 / 0.001 / 0.107 / 175.0	0.190 / 0.034 / 0.030 / 46.0	0.278 / 0.004 / 0.159 / 175.0
Diffusion H+raw	0.326 / 0.017 / 0.139 / 175.0	0.120 / 0.012 / 0.031 / 46.0	0.139 / 0.006 / 0.150 / 175.0
Diffusion H + flatline	0.350 / 0.027 / 0.151 / 115.0	0.000 / 0.000 / 0.000 / -	0.000 / 0.000 / 0.061 / -
CUSUM	0.199 / 0.016 / 0.136 / 235.0	0.014 / 0.003 / 0.083 / 190.0	0.153 / 0.000 / 0.140 / 115.0
GDN adaptation	0.546 / 0.072 / 0.243 / 55.0	0.051 / 0.006 / 0.029 / 94.0	0.157 / 0.003 / 0.190 / 175.0

quality rule does not force identical eligibility when bootstrap donors have additional missing values.

For synthetic, the paired entropy-minus-synchronization recall difference is 0.011 [-0.057, 0.080], and its IoU difference is -0.004 [-0.009, 0.001]. Entropy misses 437 event-seed pairs; conditional detected-event IoU is 0.054, versus 0.017 when misses count as zero. The mean oracle candidate-group IoU is 0.291.

For Intel, the paired entropy-minus-synchronization recall difference is -0.144 [-0.283, 0.000], and its IoU difference is -0.022 [-0.048, -0.002]. Entropy misses 193 event-seed pairs; conditional detected-event IoU is 0.084, versus 0.009 when misses count as zero. The mean oracle candidate-group IoU is 0.429.

For PEMS, the paired entropy-minus-synchronization recall difference is -0.042 [-0.120, 0.019], and its IoU difference is 0.001 [-0.006, 0.010]. Entropy misses 186 event-seed pairs; conditional detected-event IoU is 0.042, versus 0.006 when misses count as zero. The mean oracle candidate-group IoU is 0.342.

The comparison is against specific scoring rules. Entropy cannot add information to the full correlation matrix from which it is computed; an empirical difference from the Frobenius baseline does not contradict this fact. Type macros, seed variability, event precision, sensor-level precision/recall/F1, topology/severity/duration strata and every raw PR curve are retained in the artifact.

B. Reference fidelity and calibration

Table III tests distributions, not just forecast means. The raw energy score uses independent ensemble pairs and a Euclidean norm normalized by the square root of the jointly observed dimension count. Comparing references uses their common support; an empty support has no score. Covariance errors use training-scaled units. Entropy/change coverage, interval width, correlation/covariance errors and raw energy scores are saved separately, with their support counts. These compact forecasts are model-based references, not known healthy trajectories.

TABLE III

DEFINED VALUES FROM SIX FIXED UNTOUCHED UNITS AND THREE SEEDS. RAW/H COV. ARE 90% INTERVAL COVERAGES ON IDENTICAL SUPPORT FOR BOTH REFERENCES; R ERROR IS MEAN ELEMENT RMSE. ENERGY/COVARIANCE ERRORS AND SUPPORT COUNTS ARE RETAINED. UNDEFINED COMMON SUPPORT IS SHOWN AS -.

Data	Reference	Raw cov.	H cov.	R error
Syn. 64	Block	0.880	0.899	0.239
Syn. 64	Diffusion	0.972	0.372	0.322
Intel	Block	0.571	0.898	0.386
Intel	Diffusion	0.650	0.092	0.481
PEMS	Block	0.837	0.872	0.270
PEMS	Diffusion	0.869	0.521	0.369

Raw marginal coverage does not ensure organizational fidelity: synthetic diffusion raw coverage is 0.972, but H coverage is 0.372. This limits the reference model, but model error alone does not invalidate the rank theorem: that result permits any fixed scoring rule under exchangeability. Distribution differences across calibration and test units must be considered separately.

Calibration has 32 units for synthetic, 10 units for Intel, 31 units for PEMS; Intel therefore cannot attain $\alpha = .05$. For diffusion entropy at .10, untouched per-issuance exceedance is 0.240/0.308/0.032, while legitimate-transition exceedance is 0.170/0.301/0.167. These are empirical background rates. Native recordings lack adjudicated failure labels, and comparable nominal thresholds do not mean matched realized false-alert rates. Thus a recall advantage across reference models alone would not demonstrate superiority at equal false-alarm rates.

A separate 20,000-replicate diagnostic fixes the score to the maximum absolute value of four stationary AR(1) sensors. At nominal .10 with 32 calibration units, exceedance is 0.094 for IID units and 0.164 at autocorrelation .99. Spacing by 32 steps gives 0.100 for this specified process; it is not a validity theorem for gapped real data.

Table IV is a post-primary diagnostic at common empirical budgets. For each method/configuration, the largest attainable

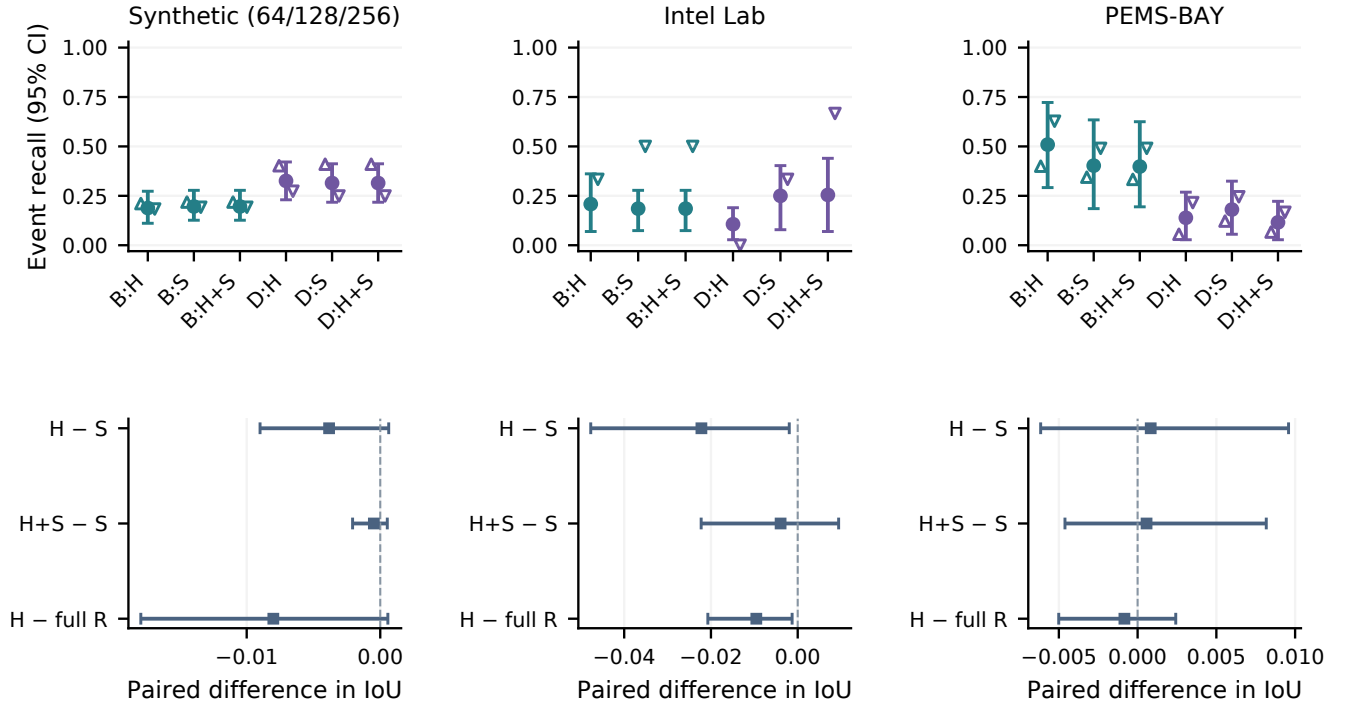


Fig. 3. At nominal $\alpha = .10$, top panels show overall event recall with whole-block 95% intervals; open up/down markers show observed-increase/decrease strata. B denotes bootstrap, D diffusion, H entropy and S synchronization. Bottom panels are paired differences in unconditional localization IoU, with 95% intervals; positive favors the left-hand feature set.

TABLE IV

RETROSPECTIVE COMPARISON AT UNTOUCHED-CONTROL ISSUANCE BUDGETS OF AT MOST .10. EACH CELL IS EVENT RECALL / UNCONDITIONAL IOU / ACHIEVED BACKGROUND EXCEEDANCE. THRESHOLDS USE REUSED TEST CONTROLS, SO THESE ARE DESCRIPTIVE OPERATING POINTS, NOT DEPLOYMENT GUARANTEES.

Method	Synthetic	Intel	PEMS
Bootstrap H	0.279 / 0.020 / 0.088	0.000 / 0.000 / 0.000	0.319 / 0.023 / 0.088
Bootstrap S	0.258 / 0.032 / 0.073	0.000 / 0.000 / 0.000	0.148 / 0.011 / 0.028
Bootstrap H+S	0.258 / 0.031 / 0.073	0.000 / 0.000 / 0.000	0.185 / 0.017 / 0.032
Diffusion H	0.037 / 0.006 / 0.011	0.000 / 0.000 / 0.000	0.250 / 0.013 / 0.092
Diffusion S	0.056 / 0.008 / 0.028	0.000 / 0.000 / 0.000	0.181 / 0.005 / 0.081
Diffusion H+S	0.056 / 0.008 / 0.028	0.000 / 0.000 / 0.000	0.144 / 0.004 / 0.066
Diffusion full R	0.165 / 0.017 / 0.038	0.000 / 0.000 / 0.000	0.162 / 0.007 / 0.058
GDN adaptation	0.460 / 0.063 / 0.074	0.000 / 0.000 / 0.000	0.176 / 0.005 / 0.068

rank threshold with untouched-control exceedance at most .10 is chosen, pooling seeds and using no injected labels. Whole-block paired resampling reselects thresholds. The same controls select and describe these points: this is retrospective comparison, not prospective false-alert validation, and finite rank resolution often leaves unused budget. Primary thresholds remain unchanged.

At this empirical budget, the paired H-minus-S recall difference for synthetic is -0.019 [-0.130, 0.034].

At this empirical budget, the paired H-minus-S recall difference for PEMS is 0.069 [-0.065, 0.236].

For Intel, the displayed methods have no nonzero attainable rank threshold within the .10 control budget; their zero recall is a resolution/shift limitation, not evidence of equal feature quality.

C. Ablations, quality and persistent history

The post-run audit finds 6/360 scheduled interventions with no modified finite value or newly removed observation, at a 10^{-6} training-scaled tolerance (6 intel matched-covariance). These remain in the primary denominators; scheduled interventions do not guarantee visible changes. Coverage and covariance guards limit the real-data benchmark.

On the identical three-fault limited subset for synthetic 64, entropy recall with B=32/64/128 is 0.000/0.000/0.000. These small subsets quantify Monte Carlo sensitivity and cost, not a precise ranking of sample budgets.

On the identical three-fault limited subset for pems, entropy recall with B=32/64/128 is 0.000/0.000/0.000. These small subsets quantify Monte Carlo sensitivity and cost, not a precise ranking of sample budgets.

TABLE V

DIFFUSION-ENTROPY ABLATION RECALL AT NOMINAL $\alpha = .10$, SEED 17 THROUGHOUT. EACH RESTRICTED FAMILY HAS ITS OWN CALIBRATION MAXIMA; THE LOCALIZATION BUDGET STAYS FIXED. LAST TWO ROWS LIST THE THREE LABELED SETTINGS IN ORDER.

Setting	Synthetic	Intel	PEMS
$H + \Delta H$	0.292	0.069	0.083
H level only	0.315	0.014	0.083
Higher than expected	0.315	0.542	0.431
Lower than expected	0.421	0.056	0.083
Actual increase only	0.324	0.153	0.181
Actual decrease only	0.296	0.097	0.083
Graph conditioning removed	0.190	0.056	0.083
Graph conditioning shuffled	0.282	0.069	0.111
Group size 6 / 12 / 24	0.264 / 0.421 / 0.486	0.069 / 0.042 / 0.000	0.167 / 0.167 / 0.250
Window 24 / 48 / 96	0.389 / 0.315 / 0.264	0.125 / 0.111 / 0.194	0.333 / 0.167 / 0.083

The global/local sensitivity uses the same $W=160$ bootstrap forecasts and calibration budget at 64 nodes. Local/global recall is 0.125/0.014, and IoU is 0.018/0.003. A global detection identifies the entire network, whereas the local arm uses its fixed sensor budget. The first valid issuance is 55 samples after onset, so short events may finish first. The global sample-support rule requires longer windows: the other configurations cannot supply enough calibration units to attain .10 under their original split/episode boundaries. This is a feasibility and longer-window comparison, not an isolated comparison with the primary shorter windows.

For synthetic 64, the eligible-group fraction falls from 1.000 to 0.000 after 10% additional entrywise missingness.

For intel, the eligible-group fraction falls from 0.068 to 0.000 after 10% additional entrywise missingness.

For pems, the eligible-group fraction falls from 0.982 to 0.000 after 10% additional entrywise missingness.

Among 54 inspected group windows, the labeled Gaussian covariance-EM sensitivity supplies a PSD estimate in 3 cases that the primary rule cannot use; it assumes a Gaussian missing-at-random model and does not inherit the detector's calibration. Separate retained-span light/voltage features and their coverage are preserved in the Intel auxiliary audit. Quality-hybrid outcomes remain separate from entropy-only results.

In the long-fault sensitivity, mean late-event entropy maxima for contaminated, paired-clean and frozen-earlier histories are 9.776/9.629/10.114. These are sensitivity scores, not newly calibrated decisions. The intervention changes the conditioning history; a historical reference need not remain useful after a persistent regime change. The unscreened-training fidelity comparison is also retained, rather than assuming the real training data are healthy.

D. Computational cost and operational delay

On the RTX A6000, complete-detector p50 latency ranges from 0.271 to 0.986 s per issuance, with p95 0.359–0.988 s and peak allocated memory 1396.8 MiB. Training, scoring and required GPU-enabled audits record 2.80 stage-hours within the four-hour ceiling. For the identical 128-window, 96-row, 24-sensor measurement batch, CPU float32/GPU float32

p50 time is 4.28; this is a kernel comparison, not an end-to-end CPU speedup claim. The Xeon Gold 6342 (nominal 2.80 GHz) uses four Torch threads, with PyTorch 2.8.0 and CUDA 12.8. Model-only and complete-pipeline throughput, p50/p95, hardware/software and sample-count tradeoffs are saved. Serial computation is added to timestamp delays in the artifact; issuance spacing and window availability dominate these sub-second/second execution costs.

VII. AUDITABLE MONITORING AND LIMITATIONS

The saved replay links a stable spatial node/edge view, ranked groups, signed residual heatmap and synchronized raw, entropy and non-entropic traces. Blue/purple down markers and orange/red up markers label actual ΔH ; higher/lower than the generated expectation is shown separately. Gray denotes inadequate support and a distinct symbol identifies flatline quality status. Inspection exposes group membership, common-row counts, units, observed and reference correlations, shared-channel evidence and its support, checkpoint, forecast issuance, observation time, window and scan-adjusted value. Physical edges and correlation overlays are distinct. Reference bands use the same samples as all compared traces. Trailing residual clustering is descriptive, uses only past/current observations, and introduces no additional significance claim. No firmware, manufacturer, maintenance or road-segment metadata are invented; spatial proximity is an input, not a discovered cause.

Several limitations matter for deployment. Spectral compression loses sign, sensor identity within a fixed group and other covariance information. Selection of groups limits localization, while common-case deletion can make a large scan ineligible. The Gaussian PSD sensitivity makes additional missingness/model assumptions and does not silently replace the primary estimator. Forecast misspecification or contaminated history can normalize persistent faults. Gapped real calibration units still exhibit dependence, and small calibration sets limit attainable tail levels. Synthetic mechanisms and real-data injections do not establish accuracy on confirmed field failures. The dashboard is an evidence-inspection artifact; no operator-usability improvement is claimed.

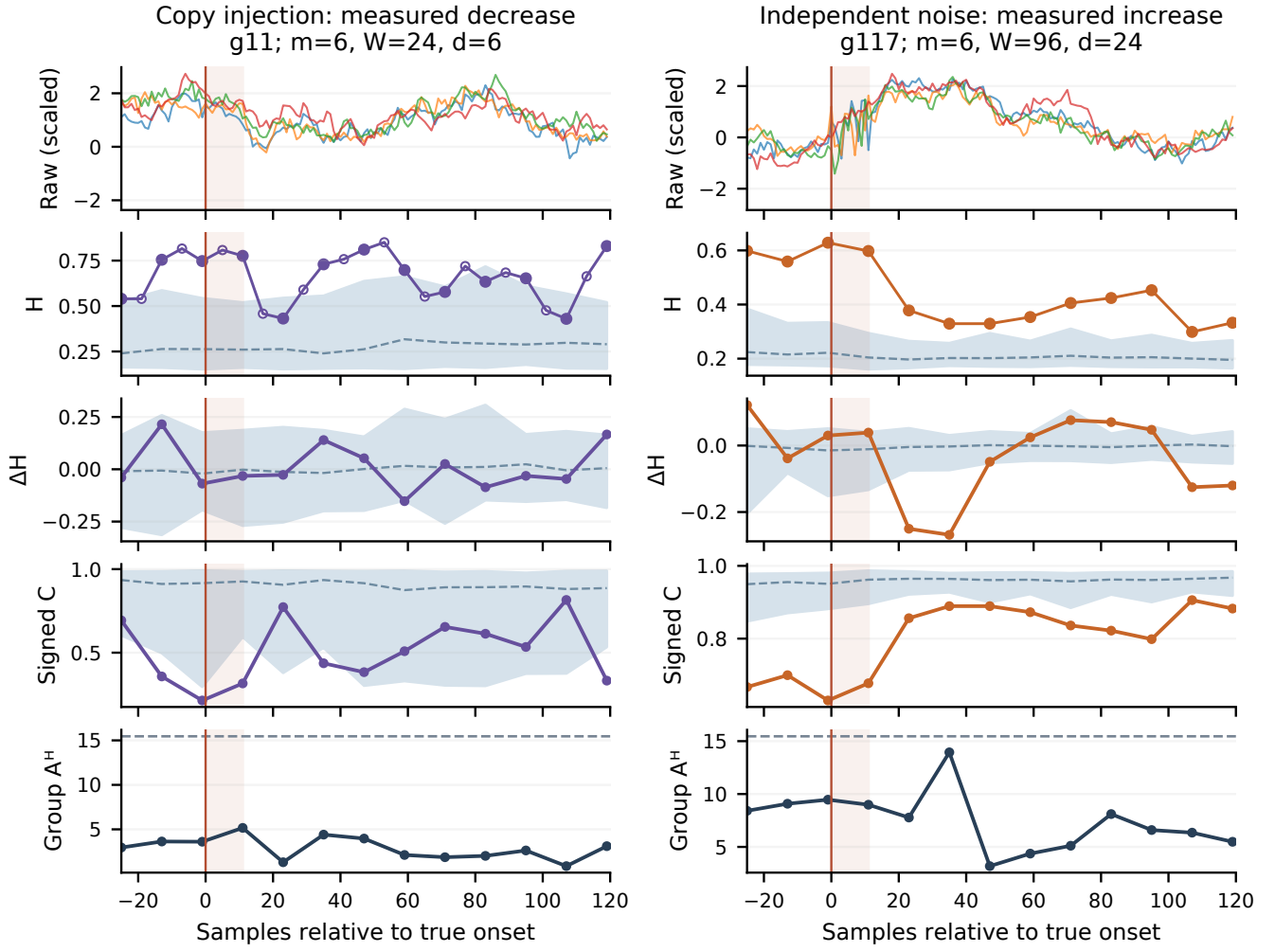


Fig. 4. Two fixed saved injections at 64 nodes, showing four raw sensor traces per group. The red span is the true 12-sample event; vertical lines mark onset. Filled H markers are issuance levels and open markers their measured lagged levels, since $W/4$ need not equal the alert stride. Reference bands/medians share the same generated blocks. The dashed score line is the strict scan threshold. Groups are selected by signed measured change among candidates overlapping the injected set, solely for illustration; neither displayed group crosses the threshold during the event.

VIII. CONCLUSION

The paired comparisons do not establish a consistent entropy advantage in both detection and localization. The exact counterexample shows that entropy can distinguish correlation structures sharing simple synchronization summaries, while its invariances prevent sign, sensor-order or causal interpretation. In this compact forecasting study, reference fidelity, calibration shift, sample support and the group library constrain practical detection. Equal nominal rank thresholds do not establish equal physical false-alarm rates. The replay exposes measurement support and separates controlled interventions from unknown native causes. Stronger predictive references and more representative calibration require prospective validation before deployment claims.

REPRODUCIBILITY AND ACKNOWLEDGMENTS

The accompanying local research artifact includes acquisition/preprocessing scripts, raw-source hashes, frozen con-

figurations, split/group manifests, training logs/checkpoint hashes, saved predictions, replay data, source-to-equation checks and figure/table generators. Three training seeds and the predeclared four-hour GPU runtime ceiling are recorded. Intel data collection is credited to Bodik, Hong, Guestrin, Madden, Paskin and Thibaux. AI assistance was used for code, analysis scripts and manuscript drafting; experimental measurements and their limitations are traceable to saved outputs.

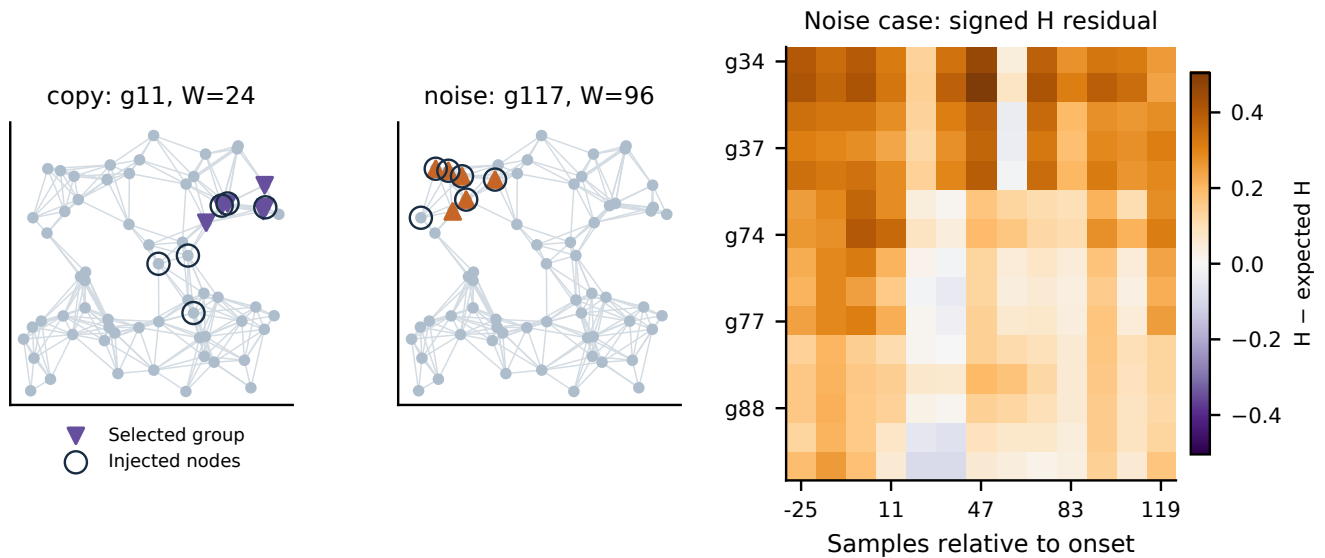


Fig. 5. Saved-case membership and signed residual traces. Outlines mark injected sensors; triangles mark the displayed group. Heatmap rows are selected by injected-set overlap for illustration. H minus expected H differs from the actual slope used for the node markers. Group locality is predefined, not a discovered cause.

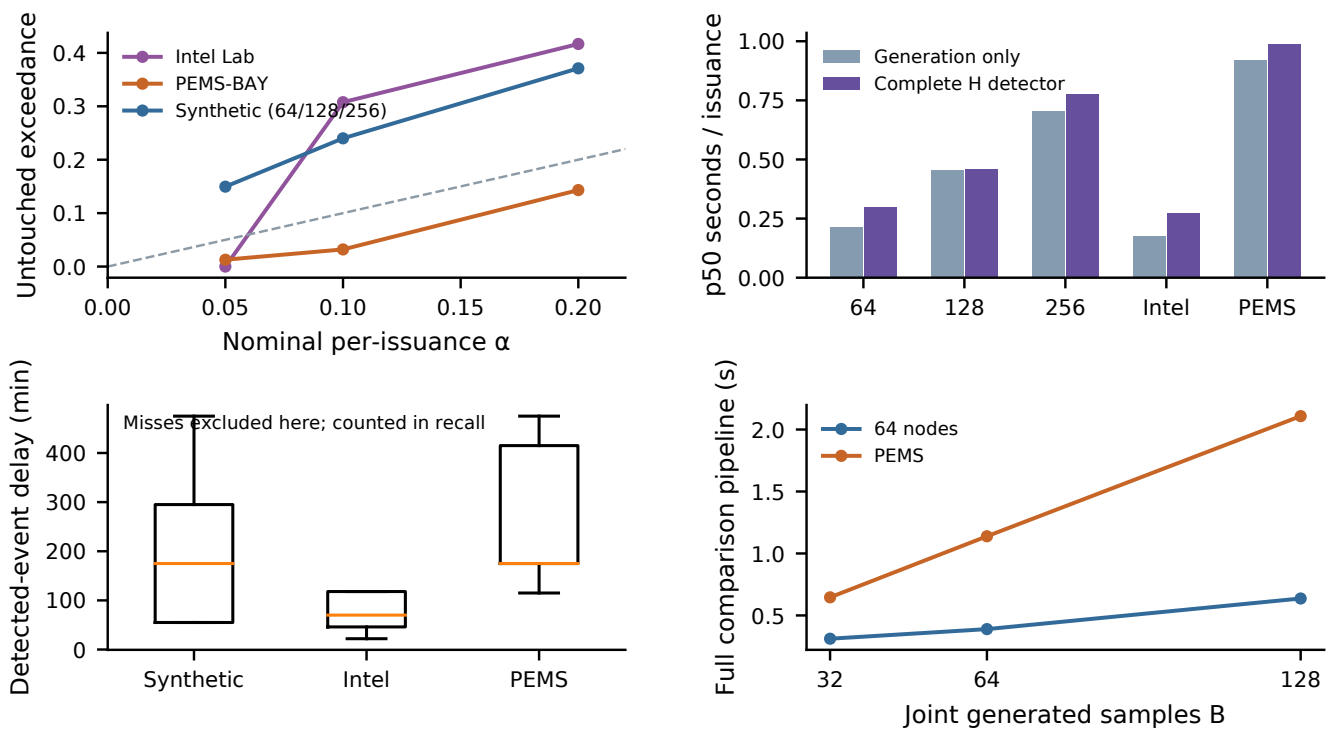


Fig. 6. Empirical untouched exceedance, synchronized execution timing, detected-event delay and generated-sample cost. Calibration is per issuance; missed events are excluded only from the delay boxplots and are counted in recall/localization. B-cost timing includes the full comparison pipeline on the same limited event subset.

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